



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

DEPARTMENT OF PHYSICS

PHYSICS LAB 6

Fall 2023-2024

LAB REPORT ON

A) Study of Ohm's law using unknown resistances.

B) Determination of the equivalent resistances for series and parallel combinations of resistors.

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Submitted By

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2	Nayem Siddiki	23-52361-2	Theory, Discussion, Reference, Calculation
3	Fahim Al Noor	23-51977-2	Analysis, Discussion, Result, Theory
4	Amanur Rahman	23-52039-2	Procedure, Discussion, Reference, Theory
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1. Theory:

Ohm's Law is a fundamental principle in electrical engineering and physics that describes the relationship between voltage, current, and resistance in an electrical circuit.

Ohm's Law states that the current passing through a conductor between two points is directly proportional to the voltage across the two points and inversely proportional to the resistance of the conductor. Mathematically, Ohm's Law is represented as:

$$V = I \times R$$

Where:

- V is the voltage across the conductor (measured in volts, V).
- I is the current flowing through the conductor (measured in amperes, A).
- R is the resistance of the conductor (measured in ohms, Ω).

In simple terms, this law states that the current through a conductor increases with an increase in voltage and decreases with an increase in resistance.

It's important to note that Ohm's Law is applicable only to passive components like resistors, and not to components like diodes and transistors, which exhibit more complex behavior. However, for many practical applications involving resistors and simple circuits, Ohm's Law is crucial and useful.

$$V = I R$$

Here,

V = Potential Difference

I = Current

R = Resistance
gives the value of R .

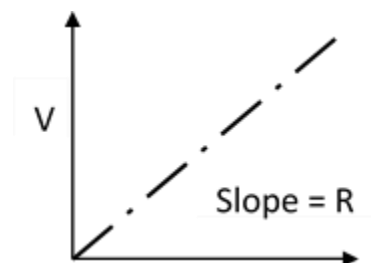


Figure – 2: Slope of the V vs I graph

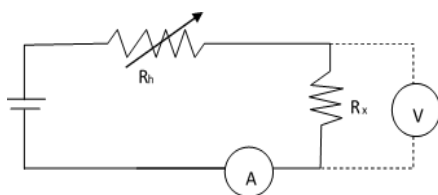


Figure – 1: Simple circuit to determine unknown resistance (R_x) by using Ohm's law. Ammeter (A) and volt meter (V) are used to measure the current and potential drop in the circuit, variable resistor, R_h is used to change the current flow in the circuit.

When N number of resistors are connected in series and parallel connections their equivalent resistances R_s and R_p are calculated by the following two equations:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N} \qquad R_s = R_1 + R_2 + \dots + R_N,$$

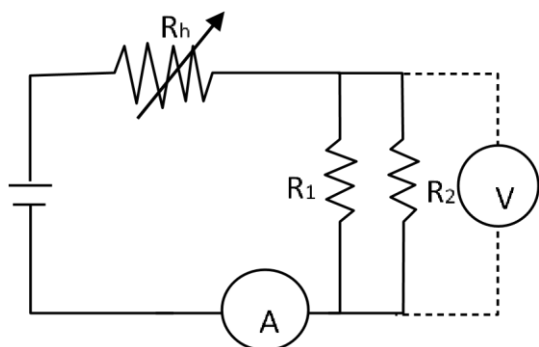


Figure – 3: Series connection for two unknown resistors R_1 & R_2 .

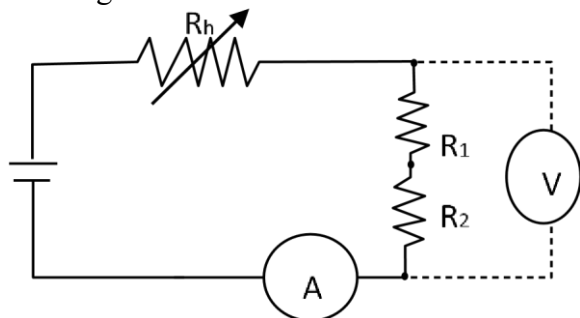


Figure – 4: Parallel connection for two unknown resistors R_1 & R_2 .

2. Apparatus:

- i. Power Supply
- ii. Variable Resistor or Rheostat
- iii. Ammeter
- iv. Voltmeter
- v. Unknown Resistors
- vi. Connecting Wires

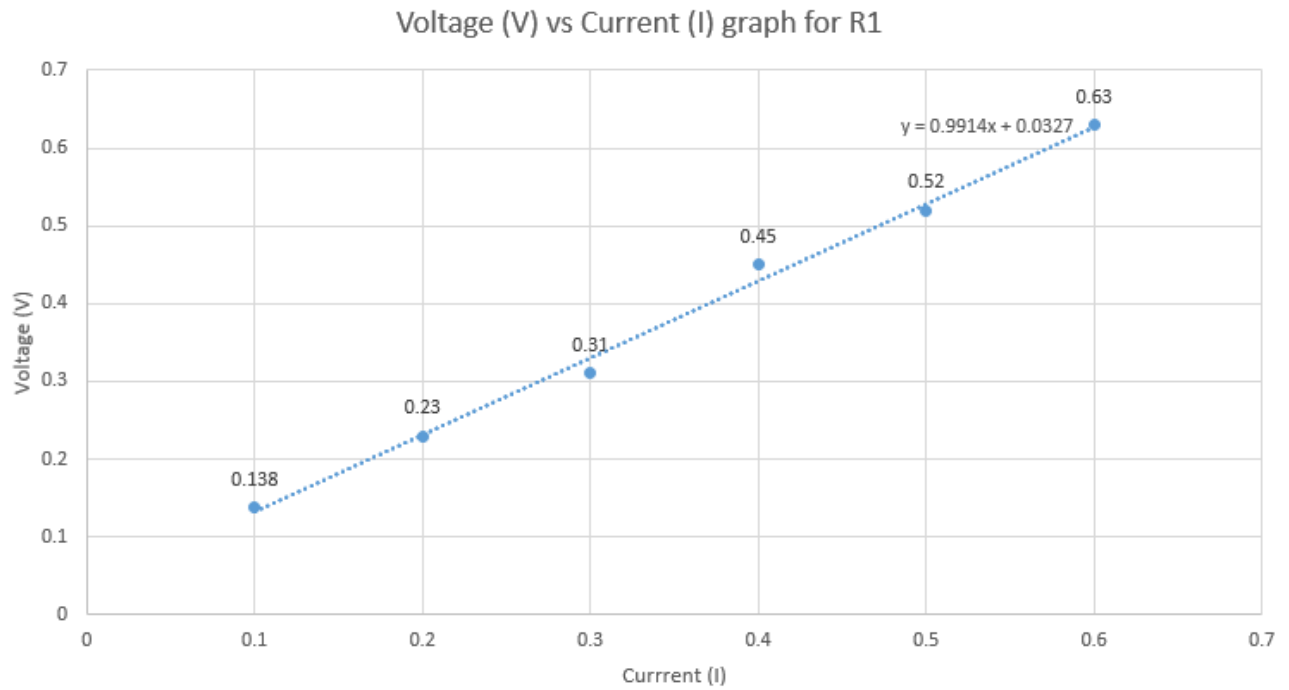
3. Procedure:

- i. Two unknown resistances, R_1 & R_2 , were used to construct the above circuits.
- ii. The rheostat (R) was employed, and a current value of less than 1 A was selected.
- iii. For six different currents, the value of (R_h) was varied, and the ammeter (A) readings were recorded.
- iv. The corresponding potential differences in the voltmeter (V) were also measured.
- v. V vs I graphs of R_1 , R_2 , R_1 & R_2 in series, and R_1 & R_2 in parallel were plotted using EXCEL.
- vi. The best fit lines were determined, and the slope of the best fit lines was calculated to find the values of R_1 , R_2 , R_p & R_s .
- vii. The values of R_p & R_s were also calculated in accordance with the laws of series and parallel combinations of resistors using the values of R_1 & R_2 .

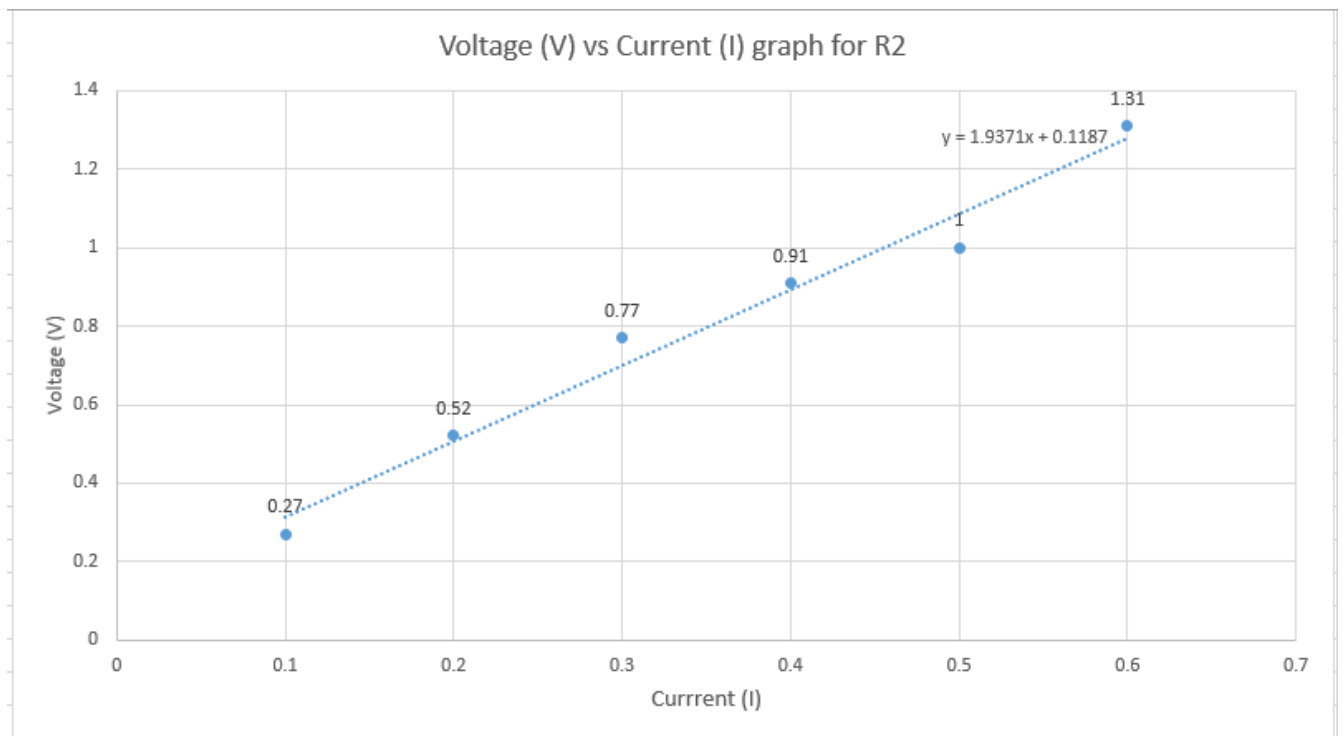
4. Experimental Data:

Table 1: Voltage and Current records for R_1 and R_2

Resistors	Current I(A)	Voltage V(V)
R1	0.1	0.138
	0.2	0.230
	0.3	0.310
	0.4	0.450
	0.5	0.520
	0.6	0.630
R2	0.1	0.270
	0.2	0.520
	0.3	0.770
	0.4	0.910
	0.5	1.000
	0.6	1.310



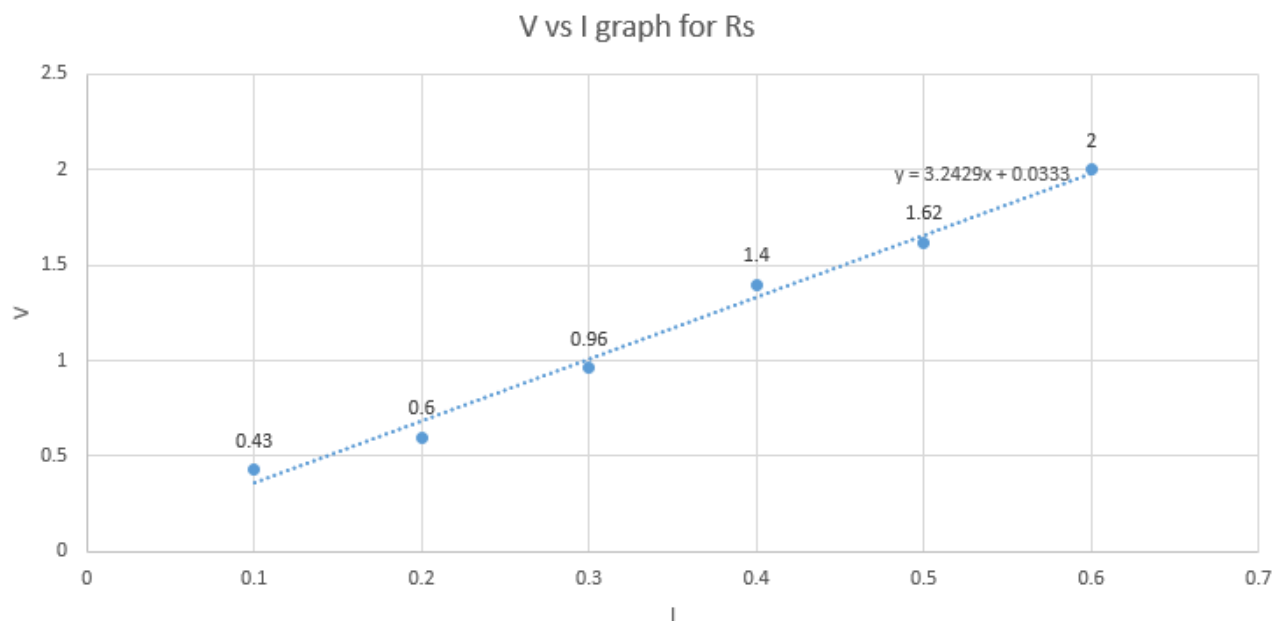
Graph 1 : Voltage (V) vs Current (I) graph for R₁



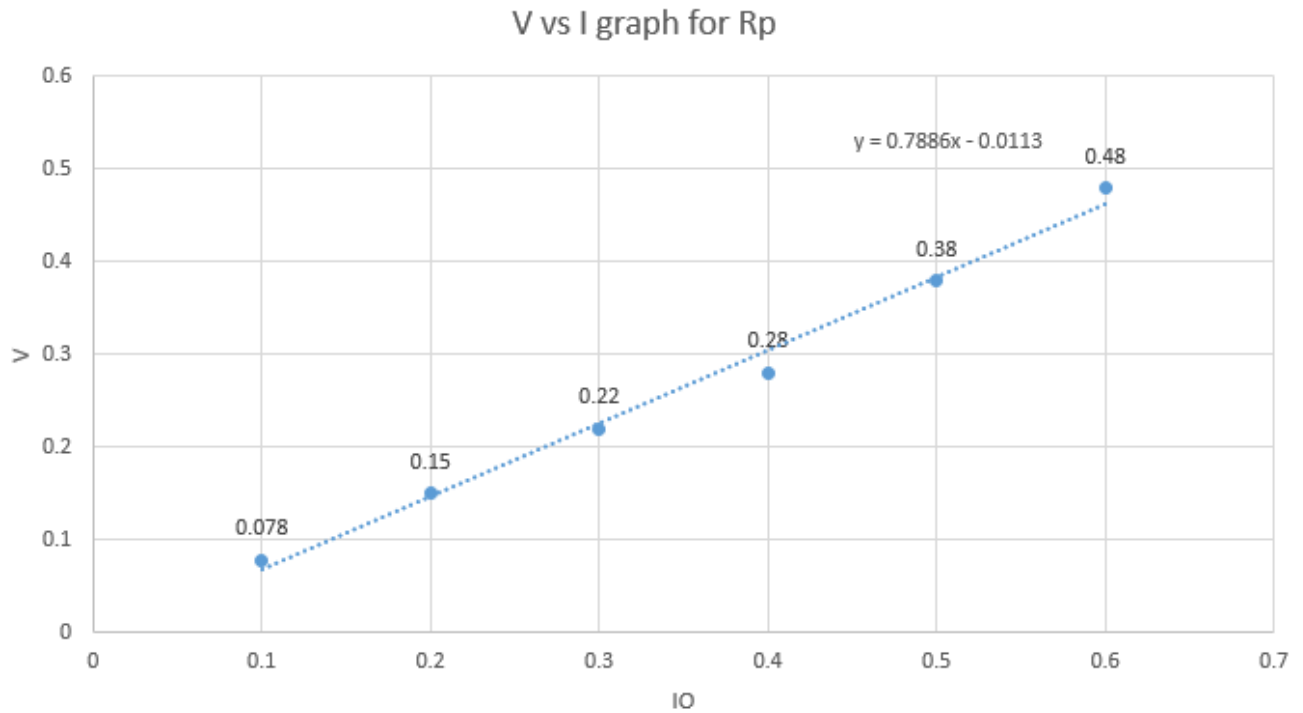
Graph 2 : Voltage (V) vs Current (I) graph for R₂

Table 2: Voltage current records for series and parallel connections.

Combinations of R_1 and R_2	Current $I(A)$	Voltage $V(V)$
Series Combination	0.1	0.430
	0.2	0.600
	0.3	0.960
	0.4	1.400
	0.5	1.620
	0.6	2.00
Parallel Combination	0.1	0.078
	0.2	0.150
	0.3	0.220
	0.4	0.280
	0.5	0.380
	0.6	0.480



Graph 3 : Voltage (V) vs Current (I) graph for R_s



Graph 4 : Voltage (V) vs Current (I) graph for R_p

5. Calculations:

- Value of R₁ from graph = 0.9914 Ω
- Value of R₂ from graph = 1.9371Ω
- **Calculated value of R₁ & R₂ in series:**

$$R_s = R_1 + R_2$$

$$\text{or, } R_s = 0.9914 + 1.9371$$

$$\text{or, } R_s = 2.9285 \Omega$$

- Calculated value of R₁ & R₂ in parallel:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\text{or, } R_p = \frac{R_1 \times R_2}{R_1 + R_2} = 0.6$$

$$\text{or, } R_p = \frac{0.9914 \times 1.9371}{0.9914 + 1.9371} = 0.655$$

6. Result:

Resistances from the graphs		Calculated Values of R_s and R_p in Ohms	Comments
Resistors	Values in Ohms		
R_1	0.9914 Ω	2.9285 Ω 0.6558 Ω	The discrepancies between the calculated value of R_s and R_p with their graphical values is a result of not having enough accuracy of the values taken when calculating results.
R_2	1.9371 Ω		
R_s	3.2429 Ω		
R_p	0.7886 Ω		

7. Discussion:

Ohm's Law states that the current passing through a conductor is directly proportional to the voltage applied across it and inversely proportional to its resistance. By using an unknown resistance, students can validate the law's accuracy and understand the relationship between voltage, current, and resistance. In this experiment, a circuit is constructed with a known voltage source, ammeter, and the unknown resistor. The voltage across the resistor and the current passing through it are measured. By plotting the current-voltage relationship, students can determine the resistance value. This process allows students to grasp the concept of resistance and how it affects the flow of current in a circuit.

8. References:

Fundamental of Physics (10th Edition): Ohm's Law (Chapter 26, page 756-759)

- Video Links:

<https://www.youtube.com/watch?v=Vh3XG77hgU4>

<https://www.youtube.com/watch?v=hxYn74Yhkok>

https://www.youtube.com/watch?v=2c1z4pLsI_w